Distributional Reinforcement Learning | Part 2

Jonas Müller

June 28, 2026

Outline

Recap: MDP and Return

We are working on a finite MDP:

$$(\mathcal{X}, \mathcal{A}, \xi_0, P_X, P_R), \qquad \pi : \mathcal{X} \to \mathcal{P}(\mathcal{A}), \qquad \gamma \in [0, 1).$$

Trajectory process:

$$X_0 \sim \xi_0, \qquad A_t \sim \pi(\cdot \mid X_t),$$

$$R_t \sim P_R(\cdot \mid X_t, A_t), \qquad X_{t+1} \sim P_X(\cdot \mid X_t, A_t), \qquad t \geq 0.$$

Discounted return:

$$G^\pi(x) = \sum_{t=0}^{\infty} \gamma^t R_t \quad \text{given } X_0 = x.$$

Recap: Value function and Bellman Operator

The value function is given by the expected return:

$$V^\pi(x) = \mathbb{E}[G^\pi(x)] = \int_{\mathbb{R}} z \, d\eta_\pi(x)(z)$$

Where $\eta_\pi(x) = \mathcal{D}(G^\pi(x))$ is the return distribution.
As we know, there is a recursive relation for the value function known as Bellman's Equation:

$$V^\pi = \mathcal{T}^\pi V^\pi.$$

Where we have the Bellman operator $\mathcal{T}^\pi : \mathbb{R}^{\mathcal{X}} \to \mathbb{R}^{\mathcal{X}}$:

$$(\mathcal{T}^\pi V)(x) = \mathbb{E}_\pi \left[R + \gamma V(X') \mid X = x \right].$$

Recap: Random-Variable Bellman Equation

With (A, R, X') sampled from $\pi(\cdot \mid x) P_R(\cdot \mid x, A) P_X(\cdot \mid x, A)$, the random-variable Bellman equation is:

$$G^\pi(x) \stackrel{D}{=} R + \gamma G^\pi(X') \mid X = x.$$

Taking laws gives the return-distribution equation:

$$\eta_\pi(x) = \mathcal{D}\left(R + \gamma G^\pi(X') \mid X = x\right).$$

This is the distributional version of $(\mathcal{T}^\pi V)(x) = \mathbb{E}_\pi[R + \gamma V(X') \mid X = x]$.

Recap: Distributional Bellman Operator

Define the bootstrap map

$$b_{r,\gamma}(z) = r + \gamma z, \qquad r, z \in \mathbb{R},$$

and for $Z' \sim \eta(x')$,

$$(b_{r,\gamma})_{\#}\eta(x') = \mathcal{D}(r + \gamma Z').$$

Distributional Bellman operator $\mathcal{T}^\pi : \mathcal{P}(\mathbb{R})^{\mathcal{X}} \to \mathcal{P}(\mathbb{R})^{\mathcal{X}}$:

$$(\mathcal{T}^\pi \eta)(x) = \mathbb{E}_\pi \left[(b_{R,\gamma})_{\#}\eta(X') \mid X = x \right].$$

Distributional Bellman equation:

$$\eta_\pi(x) = \mathbb{E}_\pi \left[(b_{R,\gamma})_{\#}\eta_\pi(X') \mid X = x \right], \qquad \eta_\pi = \mathcal{T}^\pi \eta_\pi.$$

Recap: Representations of Return Distributions

Categorical representation:

$$\eta(x) = \sum_{i=1}^{m} p_i^\theta(x)\delta_{\theta_i}, \qquad p_i^\theta(x) \geq 0, \quad \sum_{i=1}^{m} p_i^\theta(x) = 1.$$

Quantile representation:

$$\eta(x) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x)}, \qquad \tau_i = \frac{2i-1}{2m}, \qquad \theta_i(x) \approx F^{-1}_{\eta(x)}(\tau_i).$$

Focus of this Presentation

If $\mathcal{X}$ is large, then storing even a scaler, let alone a distribution per state is expensive. Hence, we look for suitable approximations.

Linear Value Function Approximation | Scalar Case

For weights $w \in \mathbb{R}^n$ approximate the value function as

$$V_w(x) = \phi(x)^\top w, \qquad \nabla_w V_w(x) = \phi(x).$$

For finite $\mathcal{X}$ and feature map ϕ, stack feature rows into $\Phi \in \mathbb{R}^{|\mathcal{X}| \times n}$:

$$V_w = \Phi w, \qquad \mathcal{F}_\phi = \{\Phi w : w \in \mathbb{R}^n\}.$$

For state-action pairs, use e.g. $\phi : \mathcal{X} \times \mathcal{A} \to \mathbb{R}^n$ via $Q_w(x, a) = \phi(x, a)^\top w$.

Idea

Instead of having a lookup table for the value of each state, we use a linear function to represent the function. This function can be queried cheaply.

Linear Value Function Approximation | Scalar Case

For weights $w \in \mathbb{R}^n$ approximate the value function as

$$V_w(x) = \phi(x)^\top w, \qquad \nabla_w V_w(x) = \phi(x).$$

For finite $\mathcal{X}$ and feature map ϕ, stack feature rows into $\Phi \in \mathbb{R}^{|\mathcal{X}| \times n}$:

$$V_w = \Phi w, \qquad \mathcal{F}_\phi = \{\Phi w : w \in \mathbb{R}^n\}.$$

For state-action pairs, use e.g. $\phi : \mathcal{X} \times \mathcal{A} \to \mathbb{R}^n$ via $Q_w(x, a) = \phi(x, a)^\top w$.

Idea

Instead of having a lookup table for the value of each state, we use a linear function to represent the function. This function can be queried cheaply.

Question: How do we get a good w?

Linear Approximation as Regression

Fitting a linear value function is just ordinary least squares:

$$V_w = \Phi w, \qquad \min_{w \in \mathbb{R}^n} \| V^\pi - V_w \|_2.$$

which we could solve by the normal equation.

However

In reinforcement learning, we might want to have a better approximation for some states than others.

Weighted Norm for Linear Approximation

For some $\xi \in \mathcal{P}(\mathcal{X})$ we define the weighted L_2 norm and diagonal weight matrix

$$\|V - V'\|_{\xi,2} = \left(\sum_{x \in \mathcal{X}} \xi(x)(V(x) - V'(x))^2 \right)^{1/2}. \qquad \Xi = \text{diag}(\xi(x) : x \in \mathcal{X}).$$

The problem we now want to solve is:

$$\min_{w \in \mathbb{R}^n} \|V^\pi - V_w\|_{\xi,2}.$$

Weighted Norm for Linear Approximation

For some $\xi \in \mathcal{P}(\mathcal{X})$ we define the weighted L_2 norm and diagonal weight matrix

$$\|V - V'\|_{\xi,2} = \left(\sum_{x \in \mathcal{X}} \xi(x)(V(x) - V'(x))^2 \right)^{1/2}. \qquad \Xi = \text{diag}(\xi(x) : x \in \mathcal{X}).$$

The problem we now want to solve is:

$$\min_{w \in \mathbb{R}^n} \|V^\pi - V_w\|_{\xi,2}.$$

Proposition 9.3 | Weighted Normal Equation

If ξ has full support and the columns of Φ are linearly independent, the unique solution is

$$w^* = (\Phi^\top \Xi \Phi)^{-1} \Phi^\top \Xi V^\pi.$$

Projection Back into the Feature Space

We have a linear approximation now, however we cant iterate the Bellman operator because we could have

$$\mathcal{T}^\pi V_w \notin \mathcal{F}_\phi$$

.

Define the weighted orthogonal projection

$$(\Pi_{\phi,\xi} V)(x) = \phi(x)^\top w^*, \qquad w^* \in \arg\min_w \| V - V_w \|_{\xi,2}.$$

Approximate dynamic programming therefore iterates

$$V_{k+1} = \Pi_{\phi,\xi} \mathcal{T}^\pi V_k.$$

- $\mathcal{T}^\pi$ performs the Bellman update.
- $\Pi_{\phi,\xi}$ removes components that the features cannot express.

Repeated projection can compound error (the diffusion effect).

When Is the Projected Operator Stable?

Let ξ_π be a fully supported steady-state distribution under π.

Projected Bellman contraction / Theorem 9.8

Since P^π and Π_{ϕ,ξ_π} are nonexpansions in $\|\cdot\|_{\xi_\pi,2}$,

$$\|\Pi_{\phi,\xi_\pi}\mathcal{T}^\pi V - \Pi_{\phi,\xi_\pi}\mathcal{T}^\pi V'\|_{\xi_\pi,2} \leq \gamma\|V - V'\|_{\xi_\pi,2}.$$

Therefore there is a unique fixed point $\widehat{V}^\pi = \Pi_{\phi,\xi_\pi}\mathcal{T}^\pi\widehat{V}^\pi$, with

$$\|\widehat{V}^\pi - V^\pi\|_{\xi_\pi,2} \leq \frac{1}{\sqrt{1-\gamma^2}}\|\Pi_{\phi,\xi_\pi}V^\pi - V^\pi\|_{\xi_\pi,2}.$$

Scalar Linear Approximation: Proof Chain

The scalar argument works because everything lives in a Hilbert space:

$$\mathbb{R}^{\mathcal{X}}, \qquad \langle V, V' \rangle_{\xi_\pi} = \sum_{x \in \mathcal{X}} \xi_\pi(x) V(x) V'(x).$$

The feature class $\mathcal{F}_\phi = \{\Phi w : w \in \mathbb{R}^n\}$ is a linear subspace.

1. Π_{ϕ,ξ_π} is an orthogonal projection, hence nonexpansive.
2. Stationarity of ξ_π makes P^π nonexpansive in $\|\cdot\|_{\xi_\pi,2}$.
3. The Bellman operator is affine with linear part γP^π.

Therefore

$$\Pi_{\phi,\xi_\pi} \mathcal{T}^\pi \quad \text{is a } \gamma\text{-contraction.}$$

Temporal-Difference Learning: Sampled Bellman Updates

Computing the Bellman iterates is expensive because it requires the expectation over the next state and reward. Instead, TD learning uses the sampled one-step Bellman target

$$r_k + \gamma V_w(x'_k).$$

The TD error is target minus current prediction:

$$\delta_k = r_k + \gamma V_w(x'_k) - V_w(x_k).$$

Equivalently, TD minimizes the squared TD error

$$L_k(w) = \frac{1}{2}\delta_k^2 = \frac{1}{2}\left(r_k + \gamma V_w(x'_k) - V_w(x_k)\right)^2.$$

In the semi-gradient step, only $V_w(x_k)$ is differentiated:

$$w \leftarrow w + \alpha_k \delta_k \nabla_w V_w(x_k).$$

For linear features $V_w(x) = \phi(x)^\top w$:

$$w \leftarrow w + \alpha_k \delta_k \phi(x_k).$$

Section 9.4: Semi-Gradient TD

The book first writes the linear TD update as

$$w \leftarrow w + \alpha_k \underbrace{\left(r_k + \gamma \phi(x_k')^\top w - \phi(x_k)^\top w\right)}_{\text{TD error}} \phi(x_k).$$

Then introduce a separate vector $\widetilde{w}$ in the target:

$$w \leftarrow w + \alpha_k \left(r_k + \gamma \phi(x_k')^\top \widetilde{w} - \phi(x_k)^\top w\right) \phi(x_k).$$

With fixed $\widetilde{w}$, this is SGD on the sample loss and its solution satisfies

$$\Phi w^* = \Pi_{\phi,\xi} \mathcal{T}^\pi V_{\widetilde{w}}.$$

Book's fixed-point statement

At the fixed point $\widehat{V}^\pi = \Phi \widehat{w}$ of the projected operator,

$$\mathbb{E}_\pi \left[\left(R + \gamma \phi(X')^\top \widehat{w} - \phi(X)^\top \widehat{w}\right) \phi(X)\right] = 0.$$

From Value function approximation to Distributional Approximation

Want to linearly approximate probability distributions. However in general

$$\eta_{w_1}(x) + \eta_{w_2}(x) \notin \mathcal{P}(\mathbb{R})$$

and since the distribution lives in $\mathcal{P}(\mathbb{R})$, we can have non finite support.

How to solve this?

Fix a finite dimensional family of distributions $\mathcal{F}_\theta \subset \mathcal{P}(\mathbb{R})$ and then linearly approximate the parameters of the distribution. "Linear" distributional approximation means that the *parameters of a probability representation* depend linearly on features ϕ.

There are therefore two approximation layers:

1. a finite probability representation (categorical or quantile),
2. linear value function approximation.

Finite Return-Distribution Families

For a probability measure $\eta \in \mathcal{P}(\mathbb{R})$, we have

$$F_\eta(t) = \eta((-\infty, t]), \qquad F_\eta^{-1}(\tau) = \inf\{z \in \mathbb{R} : F_\eta(z) \geq \tau\}.$$

We consider the finite dimensional approximations:

$$\text{categorical:} \qquad \eta(x) = \sum_{i=1}^{m} p_i(x)\delta_{\theta_i}, \qquad p_i(x) \geq 0, \quad \sum_{i=1}^{m} p_i(x) = 1,$$

$$\text{quantile:} \qquad \eta(x) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x)}, \qquad \tau_i = \frac{2i-1}{2m}, \qquad \theta_i(x) \approx F^{-1}_{\eta(x)}(\tau_i).$$

What is state-dependent?

Categorical representations fix θ_i and learn the probabilities $p_i(x)$. Quantile fixes equal probabilities $1/m$ and learns locations $\theta_i(x) \approx F^{-1}_{\eta(x)}(\tau_i)$.

Finite Return-Distribution Families

For a probability measure $\eta \in \mathcal{P}(\mathbb{R})$, we have

$$F_\eta(t) = \eta((-\infty, t]), \qquad F_\eta^{-1}(\tau) = \inf\{z \in \mathbb{R} : F_\eta(z) \geq \tau\}.$$

We consider the finite dimensional approximations:

$$\text{categorical:} \qquad \eta(x) = \sum_{i=1}^{m} p_i(x)\delta_{\theta_i}, \qquad p_i(x) \geq 0, \quad \sum_{i=1}^{m} p_i(x) = 1,$$

$$\text{quantile:} \qquad \eta(x) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x)}, \qquad \tau_i = \frac{2i-1}{2m}, \qquad \theta_i(x) \approx F_{\eta(x)}^{-1}(\tau_i).$$

What is state-dependent?

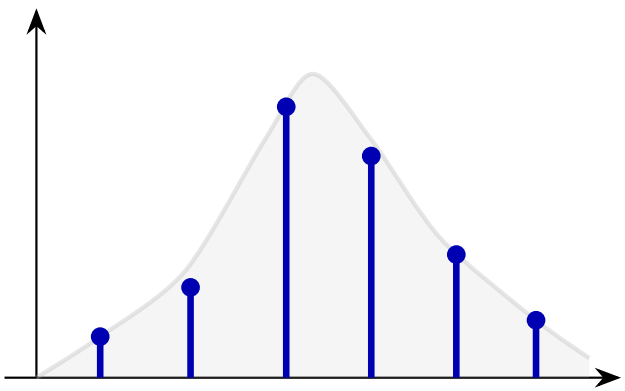
Categorical representations fix θ_i and learn the probabilities $p_i(x)$. Quantile fixes equal probabilities $1/m$ and learns locations $\theta_i(x) \approx F_{\eta(x)}^{-1}(\tau_i)$.

Question: We do not know $F_{\eta(x)}^{-1}$ in practice.

Categorical vs. Quantile

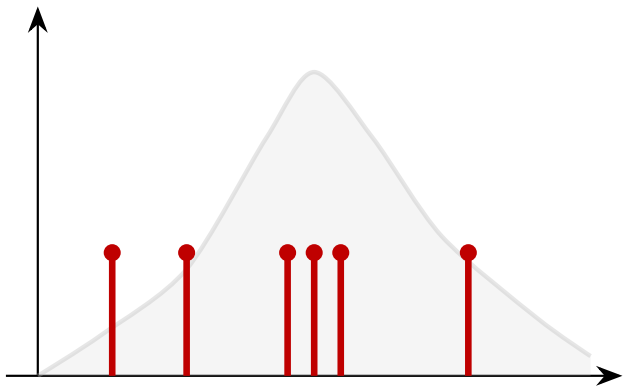
Categorical

$$\eta(x) = \sum_{i=1}^{m} p_i(x)\delta_{\theta_i}$$



Quantile

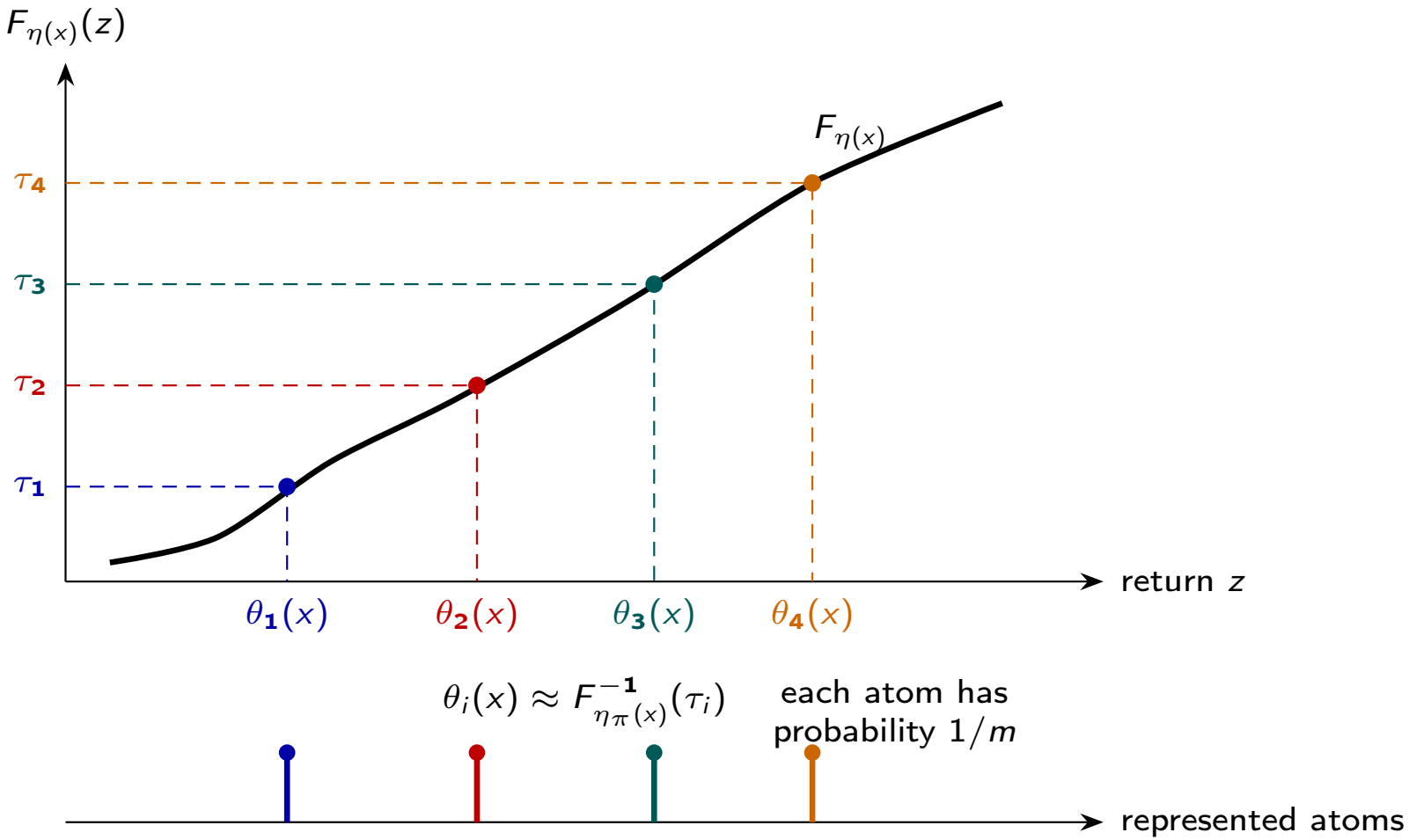
$$\eta(x) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x)}$$



Key distinction

τ_i is a fixed probability level. $\theta_i(x)$ is the return value placed at that level.

Quantile Levels: Input τ, Output θ



Changing τ_i asks for a different probability level; changing $\theta_i(x)$ moves the atom and therefore changes the represented distribution.

Distributional TD: Learning

For fixed feature map $\phi(x) \in \mathbb{R}^d$. A linear method has, for every atom,

$$u_i(x) = \phi(x)^\top w_i, \qquad w_i \in \mathbb{R}^d.$$

Observing a transition (x, r, x') gives us a empirical target. We fit this via

$$w_i \leftarrow w_i + \alpha\, \varepsilon_i(x, r, x')\, \phi(x).$$

where ε_i is a error term. This term and the meaning of u_i is what changes.

Linear Quantile TD: Representation and Loss

For the quantile representation, we choose

$$u_i(x) = \theta_i(x) = \phi(x)^\top w_i,$$

The represented return distribution is

$$\eta_w(x) = \frac{1}{m}\sum_{i=1}^{m} \delta_{\theta_i(x)} = \frac{1}{m}\sum_{i=1}^{m} \delta_{\phi(x)^\top w_i},$$

Then we take the quantile loss as in the tabular case

$$\rho_\tau(\Delta) = \left|\mathbb{1}_{\{\Delta<0\}} - \tau\right| |\Delta|.$$

For one target z, atom i uses this loss on the residual

$$L_{\tau_i}(w_i) = \rho_{\tau_i}\left(z - \phi(x)^\top w_i\right),$$

whose subgradient w.r.t. w_i is $-(\tau_i - \mathbb{1}_{\{z<\phi(x)^\top w_i\}})\phi(x)$.

Linear QTD: Distributional Semi-Gradient

Similar to the tabular case, for transition (x, a, r, x') we get m target samples

$$g_j = r + \gamma \phi(x')^\top w_j, \qquad j = 1, \dots, m.$$

Averaging the quantile subgradient over these targets yields

$$w_i \leftarrow w_i + \alpha \frac{1}{m} \sum_{j=1}^{m} \underbrace{\left(\tau_i - \mathbb{1}_{\{g_j < \phi(x)^\top w_i\}}\right)}_{\text{quantile TD error}} \phi(x).$$

Unlike the scalar TD error, each averaged quantile error is in $[-1, 1]$.

Linear Categorical TD: Prediction

We approximate the probabilites $p_i(x; w)$ by a linear combination of features:

$$p_i(x; w) = \frac{\exp(\phi(x)^\top w_i)}{\sum_{j=1}^m \exp(\phi(x)^\top w_j)}, \qquad \eta_w(x) = \sum_{i=1}^m p_i(x; w)\delta_{\theta_i}.$$

transformed into probabilities by the softmax function. Softmax is needed, direct linear coefficients need not be nonnegative or sum to one.

Linear Categorical TD: Prediction

We approximate the probabilites $p_i(x; w)$ by a linear combination of features:

$$p_i(x; w) = \frac{\exp(\phi(x)^\top w_i)}{\sum_{j=1}^m \exp(\phi(x)^\top w_j)}, \qquad \eta_w(x) = \sum_{i=1}^m p_i(x; w)\delta_{\theta_i}.$$

transformed into probabilities by the softmax function. Softmax is needed, direct linear coefficients need not be nonnegative or sum to one.

This can be handled, later!

Linear CTD: How to Stay on Fixed Support

For a sampled transition (x, r, x'), the Bellman update is an update of the prediction at the current state x:

$$\underbrace{\eta_w(x) = \sum_{j=1}^{m} p_j(x; w)\delta_{\theta_j}}_{\text{prediction at } x} \xrightarrow{\text{Observed } (r,x')} \underbrace{(b_{r,\gamma})_{\#}\eta_w(x') = \sum_{j=1}^{m} p_j(x'; w)\delta_{r+\gamma\theta_j}}_{\text{Empirical Bellman step for } x}.$$

Since in general $r + \gamma\theta_j$ need not lie on the fixed support, use the categorical projection Π_c to project the shifted atoms back to the fixed support $S = \{\theta_1, \ldots, \theta_m\}$. For each j, if $\theta_\ell \le r + \gamma\theta_j \le \theta_{\ell+1}$, split its mass as

$$\bar{p}_\ell \mathrel{+}= p_j(x'; w)\frac{\theta_{\ell+1} - (r + \gamma\theta_j)}{\theta_{\ell+1} - \theta_\ell}, \qquad \bar{p}_{\ell+1} \mathrel{+}= p_j(x'; w)\frac{(r + \gamma\theta_j) - \theta_\ell}{\theta_{\ell+1} - \theta_\ell}.$$

This gives the projected target

$$\bar{\eta}_{x,r,x'} = \Pi_c(b_{r,\gamma})_{\#}\eta_w(x') = \sum_{i=1}^{m} \bar{p}_i\delta_{\theta_i}.$$

Linear CTD: Target, Loss, and Update

To learn softmax probabilities, we use the Cross-entropy loss:

$$L(w) = -\sum_{i=1}^{m} \bar{p}_i \log p_i(x; w).$$

Since

$$\frac{\partial L}{\partial u_i} = p_i(x; w) - \bar{p}_i, \qquad \nabla_{w_i} u_i(x) = \phi(x),$$

gradient descent gives the semi-gradient CTD update:

$$w_i \leftarrow w_i + \alpha \underbrace{(\bar{p}_i - p_i(x; w))}_{\text{CTD error}} \phi(x).$$

Chapter 9 Objects: Signed Categorical Family

We can make our lives easier by allowing negative measures. The signed categorical family on $S = \{\theta_1, \ldots, \theta_m\}$ is

$$\mathcal{F}_{S,m} = \left\{ \sum_{i=1}^{m} p_i \delta_{\theta_i} : p_i \in \mathbb{R}, \sum_{i=1}^{m} p_i = 1 \right\} \subseteq \mathcal{M}(\mathbb{R}).$$

Using this we can define the signed categorical return functions

$$\mathcal{F}_{S,m}^{\mathcal{X}} = \{\eta : \mathcal{X} \to \mathcal{F}_{S,m}\}.$$

Chapter 9 Objects: Signed Categorical Family

We can make our lives easier by allowing negative measures. The signed categorical family on $S = \{\theta_1, \ldots, \theta_m\}$ is

$$\mathcal{F}_{S,m} = \left\{ \sum_{i=1}^{m} p_i \delta_{\theta_i} : p_i \in \mathbb{R}, \sum_{i=1}^{m} p_i = 1 \right\} \subseteq \mathcal{M}(\mathbb{R}).$$

Using this we can define the signed categorical return functions

$$\mathcal{F}_{S,m}^{\mathcal{X}} = \{\eta : \mathcal{X} \to \mathcal{F}_{S,m}\}.$$

How to linearly approximate this?

Chapter 9 Objects: Linear Signed Approximation

We just do a linear approximation, and force it to add to one.

$$p_i(x) = \frac{1}{m} + \phi(x)^\top w_i - \frac{1}{m}\sum_{j=1}^{m} \phi(x)^\top w_j.$$

Then

$$\sum_{i=1}^{m} p_i(x) = 1, \qquad p_i(x) \in \mathbb{R}.$$

The resulting family of distribution is denoted by $\mathcal{F}_{\phi,S,m}$.

Chapter 9 Objects: Cramér Geometry

For a signed distribution ν, let

$$F_\nu(z) = \nu((-\infty, z])$$

denote its cumulative distribution function. The Cramér distance is

$$\ell_2^2(\nu, \nu') = \int_{\mathbb{R}} (F_\nu(z) - F_{\nu'}(z))^2 \, dz$$

whenever this quantity is finite. Define

$$\mathcal{M}_{\ell_2,1}(\mathbb{R})$$

as the set of signed distributions ν with total mass $\nu(\mathbb{R}) = 1$ and finite Cramér distance to the reference distribution δ_0. For return-distribution functions $\eta : \mathcal{X} \to \mathcal{M}_{\ell_2,1}(\mathbb{R})$,

$$\ell_{\xi_\pi,2}^2(\eta, \eta') = \sum_{x \in \mathcal{X}} \xi_\pi(x) \ell_2^2(\eta(x), \eta'(x)) \,.$$

Chapter 9 Objects: The Two Projections

The categorical projection maps a signed distribution to the fixed support:

$$\Pi_c : \mathcal{M}_{\ell_2,1}(\mathbb{R}) \to \mathcal{F}_{S,m}.$$

The feature projection maps signed categorical return functions back into the linearly representable family:

$$\Pi_{\phi,\xi_\pi,\ell_2} : \mathcal{F}_{S,m}^{\mathcal{X}} \to \mathcal{F}_{\phi,S,m}.$$

It is defined as the $\ell_{\xi_\pi,2}$-closest element of the linear feature class. The doubly projected Bellman operator is

$$\Pi_{\phi,\xi_\pi,\ell_2} \Pi_c \mathcal{T}^\pi.$$

The order is: Bellman update, categorical projection, feature projection.

Chapter 9 Lemmas: Bellman and Categorical Steps

Assume rewards have finite first moments and the Markov chain under π has a unique stationary distribution ξ_π.

Lemma 9.14: Bellman Step

For any $\eta, \eta' \in \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}}$,

$$\ell_{\xi_\pi,2}(\mathcal{T}^\pi \eta, \mathcal{T}^\pi \eta') \leq \gamma^{1/2} \ell_{\xi_\pi,2}(\eta, \eta').$$

Lemma 9.15: Categorical Projection

For any signed distributions ν, ν',

$$\ell_2(\Pi_c \nu, \Pi_c \nu') \leq \ell_2(\nu, \nu').$$

Chapter 9 Lemmas: Feature Projection

The final projection is the analogue of the scalar linear projection, but applied to signed categorical distributions with the Cramér metric.

Lemma 9.16: Linear Feature Projection

For any $\eta, \eta' \in \mathcal{F}^{\mathcal{X}}_{S,m}$,

$$\ell_{\xi_\pi,2}\left(\Pi_{\phi,\xi_\pi,\ell_2}\eta, \Pi_{\phi,\xi_\pi,\ell_2}\eta'\right) \leq \ell_{\xi_\pi,2}(\eta,\eta').$$

Thus each piece is either a contraction or a nonexpansion:

$$\mathcal{T}^\pi \quad \text{contracts,} \qquad \Pi_c,\ \Pi_{\psi,\xi_\pi,\ell_2} \quad \text{do not expand.}$$

Chapter 9 Theorem: Doubly Projected Bellman

Composing the three preceding statements gives the actual convergence-style result for linear distributional approximation in the book.

Theorem 9.13

Under the assumptions above, for all $\eta, \eta' \in \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}}$,

$$\ell_{\xi_\pi,2}\left(\Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta, \Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta'\right) \leq \gamma^{1/2}\ell_{\xi_\pi,2}(\eta,\eta').$$

Hence the projected distributional Bellman operator has a unique fixed point in $\mathcal{F}_{\psi,S,m}$, by Banach's fixed-point theorem.

Chapter 9: Exact Contraction Chain

For $\eta, \eta' \in \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}}$,

$$\begin{aligned}
&\ell_{\xi_\pi,2}\left(\Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta, \Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta'\right) \\
&\quad\leq \ell_{\xi_\pi,2}(\Pi_c\mathcal{T}^\pi\eta, \Pi_c\mathcal{T}^\pi\eta') \\
&\quad\leq \ell_{\xi_\pi,2}(\mathcal{T}^\pi\eta, \mathcal{T}^\pi\eta') \\
&\quad\leq \gamma^{1/2}\ell_{\xi_\pi,2}(\eta, \eta').
\end{aligned}$$

The three lines are exactly: feature projection nonexpansion, categorical projection nonexpansion, and distributional Bellman contraction.

Why Signed Measures Make the Theory Work

The scalar proof needs a linear subspace and an orthogonal projection. The signed categorical construction recreates that geometry:

$$p_i(x) \in \mathbb{R}, \qquad \sum_i p_i(x) = 1.$$

Allowing negative masses is mathematically useful because affine combinations remain inside the signed unit-mass family.

Practical probability models lose this geometry

Softmax CTD/C51 enforces

$$p_i(x) \geq 0, \qquad \sum_i p_i(x) = 1,$$

producing a nonlinear simplex-valued parameterization, not a linear projection space.

Quantile methods move atom locations instead of masses, so their projection is quantile regression, not an orthogonal projection in Cramér geometry.

Chapter 10: Why This Is Not a DQN Theorem

C51, QR-DQN, and IQN combine several ingredients:

$$\text{nonlinear network} + \text{bootstrapping} + \text{greedy control} + \text{replay/target networks}.$$

The theory is therefore much weaker than for policy evaluation.

What Theorem 9.13 needs

A fixed policy, linear projection in a fixed metric, and a fixed operator

$$\Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi.$$

What Chapter 10 uses

Nonlinear parameters and a moving control target, e.g.

$$r + \gamma Z_{\widetilde{w}}\left(x', \arg\max_{a'} \mathbb{E}[Z_{\widetilde{w}}(x', a')]\right).$$

So Chapter 10 inherits the projection intuition, but not a comparable general convergence theorem for deep distributional DQN.

From Fixed Features to Learned Features

A deep network replaces the hand-designed map $\phi(x)$ by a learned representation. Its penultimate layer can still be read as

$$\phi_w(x) \in \mathbb{R}^n,$$

followed by linear prediction heads.

- Earlier layers learn which state distinctions matter.
- Final layers map those features to values or distribution parameters.
- Backpropagation updates both simultaneously.

Linear approximation is therefore not discarded; it remains the final layer and the conceptual model for the deep semi-gradient update.

DQN: Prediction, Behavior, and Experience

DQN maps an observation x to one value per action:

$$Q_w(x, a), \qquad a \in \mathcal{A}.$$

In the canonical Atari setup:

- four preprocessed 84×84 frames form the network input;
- convolutional layers and a fully connected layer produce 512 features;
- a linear output layer produces $|\mathcal{A}|$ action values;
- an ε-greedy policy acts from Q_w;
- transitions are stored and resampled from replay memory.

DQN: Semi-Gradient Learning

With online weights w, target weights $\widetilde{w}$, and transition (x, a, r, x'), define

$$y = r + \gamma \max_{a' \in \mathcal{A}} Q_{\widetilde{w}}(x', a').$$

The squared loss and semi-gradient update are

$$L(w) = (y - Q_w(x, a))^2,$$

$$w \leftarrow w + \alpha(y - Q_w(x, a))\nabla_w Q_w(x, a).$$

If $Q_w(x, a) = \phi(x, a)^\top w$, then $\nabla_w Q_w = \phi$: this is exactly the linear update. Replay reduces temporal correlation; the frozen target network slows the moving-target feedback loop.

Distributional Network Output

Deep distributional RL changes the output from one scalar per action to m distribution parameters per action:

$$x \longmapsto \mathbb{R}^{|\mathcal{A}| \times m}.$$

The interpretation determines the algorithm:

- **C51:** fixed locations, learned softmax probabilities;
- **QR-DQN:** equal probabilities, learned quantile locations;
- **IQN:** quantile level τ becomes an additional input.

Behavior is usually risk-neutral: derive $Q_w(x, a)$ from the predicted distribution and do ε-greedy action selection as in DQN.

C51: Prediction and Behavior

For fixed $\theta_1 < \cdots < \theta_m$, the network outputs logits $\varphi_i(x, a; w)$ and

$$p_i(x, a; w) = \frac{e^{\varphi_i(x,a;w)}}{\sum_{j=1}^{m} e^{\varphi_j(x,a;w)}}, \qquad \eta_w(x, a) = \sum_{i=1}^{m} p_i(x, a; w)\delta_{\theta_i}.$$

The induced action value is

$$Q_w(x, a) = \sum_{i=1}^{m} p_i(x, a; w)\theta_i.$$

Original C51 uses $m = 51$ evenly spaced locations; support choice imposes an important inductive bias because projected targets outside it are clipped.

C51: Bellman Target and Cross-Entropy

Select the target-network greedy action

$$a_{\widetilde{w}}(x') \in \arg\max_{a'} Q_{\widetilde{w}}(x', a').$$

Construct and project the target

$$\bar{\eta}(x, a) = \Pi_c(b_{r,\gamma})_{\#}\eta_{\widetilde{w}}(x', a_{\widetilde{w}}(x')) = \sum_{j=1}^{m} \bar{p}_j \delta_{\theta_j}.$$

Then minimize

$$L(w) = -\sum_{j=1}^{m} \bar{p}_j \log p_j(x, a; w).$$

This is the nonlinear analogue of linear CTD, except shared early-layer weights influence all particles and cannot be updated independently.

QR-DQN: Quantile Prediction

QR-DQN interprets the m outputs as locations of an equally weighted quantile distribution:

$$\eta_w(x, a) = \frac{1}{m} \sum_{i=1}^{m} \delta_{\theta_i(x,a;w)}, \qquad Q_w(x, a) = \frac{1}{m} \sum_{i=1}^{m} \theta_i(x, a; w).$$

With $a_{\widetilde{w}}(x')$ greedy by this mean, the pairwise residuals are

$$\Delta_{ij} = r + \gamma \theta_j(x', a_{\widetilde{w}}(x'); \widetilde{w}) - \theta_i(x, a; w).$$

Every predicted quantile is compared with every target quantile.

QR-DQN: Quantile Huber Loss

For $\tau_i = (2i-1)/(2m)$, QR-DQN minimizes

$$L(w) = \frac{1}{m} \sum_{i,j=1}^{m} \rho_{\tau_i}^{H}(\Delta_{ij}), \qquad \rho_{\tau}^{H}(u) = |\mathbb{1}_{\{u<0\}} - \tau| H_1(u),$$

where

$$H_1(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq 1, \\ |u| - \frac{1}{2}, & |u| > 1. \end{cases}$$

The quantile asymmetry places different heads at different parts of the distribution; the Huber transition gives smooth small-error gradients and limits the influence of large errors.

IQN: Make the Quantile Level an Input

C51 and QR-DQN output a fixed finite parameter vector. IQN instead models the quantile function itself:

$$(x, a, \tau) \longmapsto \theta_w(x, a, \tau) \approx F^{-1}_{\eta(x,a)}(\tau).$$

The level is encoded by

$$\varphi(\tau) = (\cos(\pi \tau i))_{i=0}^{M-1},$$

transformed by a fully connected layer, and combined multiplicatively with the state features. The final action heads satisfy

$$\theta_w(x, a, \tau) = \phi_w(x, \tau)^\top w_a.$$

IQN: Behavior and Learning

Approximate the mean with independent $\tau_i \sim \mathcal{U}(0,1)$:

$$Q_w(x,a) \approx \frac{1}{m}\sum_{i=1}^{m} \theta_w(x,a,\tau_i).$$

For independent online levels τ_i and target levels τ'_j, minimize pairwise losses of

$$r + \gamma\theta_{\widetilde{w}}(x', a_{\widetilde{w}}(x'), \tau'_j) - \theta_w(x,a,\tau_i).$$

Sampling many level pairs reduces gradient variance. IQN approximates the whole quantile function without committing to fixed output locations.

IQN Makes Risk Sensitivity Simple

Risk-neutral behavior samples $\tau \sim \mathcal{U}(0,1)$. Lower-tail CVaR at level $\bar{\tau}$ is

$$\mathrm{CVaR}_{\bar{\tau}}(G_w(x,a)) = \frac{1}{\bar{\tau}} \int_0^{\bar{\tau}} \theta_w(x,a,\tau)\, d\tau.$$

Approximate it by changing only the sampling distribution:

$$\mathrm{CVaR}_{\bar{\tau}}(G_w(x,a)) \approx \frac{1}{m} \sum_{i=1}^{m} \theta_w(x,a,\tau_i), \quad \tau_i \sim \mathcal{U}(0,\bar{\tau}).$$

Thus an implicit representation separates learning the return distribution from choosing how that distribution should guide behavior.

How Distributional Predictions Shape Features

Split the network into learned features and final prediction heads.

- DQN trains the representation to predict one conditional mean per action.
- C51 trains it through m probability errors per action.
- QR-DQN/IQN train it through many pairwise quantile residuals.

The type and number of predictions therefore change the learned state representation. This richer auxiliary prediction problem is a leading explanation for improved control performance even when actions are ultimately selected by the mean.

Takeaways

1. Linear approximation introduces state aliasing and a weighted projection problem; the sampling distribution is central to stability.
2. Semi-gradient TD tracks a projected Bellman process, and target weights make that relationship explicit.
3. Linear QTD and CTD make distribution *parameters* linear in features.
4. Deep methods learn the features and retain the same prediction–target–semi-gradient structure.
5. C51, QR-DQN, and IQN differ mainly in how they parameterize and train the output distribution; that choice also shapes the learned representation.

References

- Bellemare, Dabney, Rowland (2023): *Distributional Reinforcement Learning*, Chapters 9–10.
- Bellemare, Dabney, Munos (2017): A Distributional Perspective on Reinforcement Learning.
- Dabney et al. (2018): Distributional Reinforcement Learning with Quantile Regression.
- Dabney et al. (2018): Implicit Quantile Networks for Distributional Reinforcement Learning.